HTTP Status Codes in ASP.NET Core Web API

What are HTTP Status Codes?

HTTP status codes are an essential part of any Web API (RESTful Services). They provide information about the status of the HTTP request from the server to the client (browser, mobile app, Postman, etc.).

In simple terms: They tell the client what happened to their request – success, failure, or something in between.

Five Main Categories (Parent Groups)

- **1xx (Informational):** Request received, continuing
- **2xx (Successful):** Request received, understood, accepted
- **3xx (Redirection):** Further action needed
- **4xx (Client Error):** Error in request sent by client
- **5xx (Server Error):** Server failed to fulfil valid request

<u>Category 1: 1xx (Informational)</u>

Meaning: The request was received and the server is continuing to process it. Not widely used in most client-server interactions.

Status Code: 100

- **Name:** Continue
- **Meaning:** Initial part of request received, client should continue sending the rest
- **When to Use:** Large request bodies (rare in Web APIs)

Status Code: 101

- **Name:** Switching Protocols
- **Meaning:** Server is switching to a different protocol as requested
- **When to Use:** WebSocket upgrades

Category 2: 2xx (Successful)

Meaning: The client's request was successfully received, understood, and accepted by the server.

Status Code: 200

- **Name:** OK
- **Meaning:** Request succeeded. Response includes requested data.
- **When to Use:** GET (fetching data), PUT (updating), successful POST
- **ASP.NET Core Helper:** Ok(data)

Status Code: 201

- **Name:** Created
- **Meaning:** Request succeeded and a new was created
- **When to Use:** POST (creating new record). Should include Location header
- **ASP.NET Core Helper:** CreatedAtAction()

Status Code: 202

- **Name:** Accepted
- **Meaning:** Request accepted but processing may not be complete immediately
- **When to Use:** Asynchronous operations (background jobs, batch processing)
- **ASP.NET Core Helper:** Accepted()

Status Code: 204

- **Name:** No Content
- **Meaning:** Request succeeded but no content to return (response body empty)
- **When to Use:** DELETE operation, or PUT with no response body
- **ASP.NET Core Helper:** NoContent()

Category 3: 3xx (Redirection)

Meaning: Further actions need to be taken to complete the request, usually a subsequent request to a different URL.

Status Code: 301

- **Name:** Moved Permanently
- **Meaning:** Resource moved permanently to a new URL. Client should use new URL for all future requests
- **When to Use:** API version changes (v1 → v2), URL structure changes
- **ASP.NET Core Helper:** RedirectPermanent()

Status Code: 302

- **Name:** Found (Temporary Redirect)
- **Meaning:** Resource temporarily located at a different URL. Client should continue using original URL
- **When to Use:** Temporary maintenance page, A/B testing
- **ASP.NET Core Helper:** Redirect()

Status Code: 304

- **Name:** Not Modified
- **Meaning:** Resource has not changed since last request. Client can use cached version
- **When to Use:** Conditional GET requests (ETag, Last-Modified)
- **ASP.NET Core Helper:** Not a direct helper (requires manual implementation)

Category 4: 4xx (Client Error)

Meaning: There was an error on the client's side – bad syntax, invalid data, missing authentication, or requesting non-existent resource.

Status Code: 400

- **Name:** Bad Request
- **Meaning:** Server cannot process due to client error (malformed syntax, invalid data)
- **When to Use:** Missing required field, invalid data type, validation failure
- **ASP.NET Core Helper:** BadRequest("error message")

Status Code: 401

- **Name:** Unauthorized
- **Meaning:** Authentication required, failed, or not provided
- **When to Use:** Missing API Key, invalid token, expired credentials
- **ASP.NET Core Helper:** Unauthorized()

Status Code: 403

- **Name:** Forbidden
- **Meaning:** Authenticated but not authorized to access resource
- **When to Use:** Valid user but no permission (regular user trying to access admin endpoint)
- **ASP.NET Core Helper:** Forbid()

Status Code: 404

- **Name:** Not Found
- **Meaning:** Requested resource does not exist on server
- **When to Use:** Wrong URL, deleted resource, invalid ID
- **ASP.NET Core Helper:** NotFound("message")

Status Code: 405

- **Name:** Method Not Allowed
- **Meaning:** HTTP method not supported by target resource
- **When to Use:** POST endpoint called with GET verb

- **ASP.NET Core Helper:** Automatic (framework returns when route exists with different verb)

<u>Category 5: 5xx (Server Error)</u>

Meaning: The server failed to fulfil a valid request. The problem is on the server's end, not the client's request.

Status Code: 500

- **Name:** Internal Server Error
- **Meaning:** Unexpected condition prevented server from fulfilling request
- **When to Use:** Unhandled exception, database connection failure, null reference
- **ASP.NET Core Helper:** Automatic (framework returns for unhandled exceptions). **Manual:** StatusCode(500)

Status Code: 501

- **Name:** Not implemented
- **Meaning:** Server does not recognize request method or cannot fulfil it
- **When to Use:** Feature not implemented yet in API
- **ASP.NET Core Helper:** StatusCode(501)

Status Code: 502

- **Name:** Bad Gateway
- **Meaning:** Server received invalid response from upstream server
- **When to Use:** API gateway failing to reach backend service
- **ASP.NET Core Helper:** Not directly returned (gateway/proxy layer)

Status Code: 503

- **Name:** Service Unavailable
- **Meaning:** Server not ready to handle request (maintenance or overload)
- **When to Use:** Scheduled maintenance, server under heavy load, rate limiting exceeded
- **ASP.NET Core Helper:** StatusCode(503)

Status Code: 504

- **Name:** Gateway Timeout
- **Meaning:** Server did not receive timely response from upstream server
- **When to Use:** Backend service taking too long to respond
- **ASP.NET Core Helper:** Not directly returned (gateway/proxy layer)

ASP.NET Core Helper Methods Quick Reference

Status Code	Helper Method
200	Ok(data)
201	CreatedAtAction()
202	Accepted()
204	NoContent()
301	RedirectPermanent()
302	Redirect()
400	BadRequest()
401	Unauthorized()
403	Forbid()
404	NotFound()
405	Automatic
500	Automatic/ StatusCode(500)
503	StatusCode(503)

Why HTTP Status Codes are Important

- **Communication Clarity:** Provides essential feedback about request result
- **Efficient Troubleshooting:** Helps identify and diagnose issues
- **Better User Experience:** Provides meaningful feedback to users
- **System Monitoring:** Helps analyze traffic pattern and optimize performance
- **Security:** Manages access control and protects resources